

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Artificial Intelligence and Data Science

ASSESSMENT TOOL 2 – Industry Problem Solving Task

Course Code:	DSA03	Course Title:	Natural Language Processing
CO Assessed:	CO4 – Precision	Assessment:	Assessment Tool 2 – Industry Problem Solving Task
Weightage:	35% of CIA		
CO5 Statement:	Formulate context-free grammars, feature structures, probabilistic parsing techniques and to construct parsing frameworks. (BL-4)		
Student Name:		Reg. No.:	
Section / Batch:		Date:	

Code of Conduct

I B V Chaitanya Reddy-192425376 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: _chaitanya_____

Instructions

- Read each industry scenario carefully before answering.
- Analyze the given problem from a semantic analysis perspective.
- Provide structured and logical answers with proper justification.
- Support your analysis using examples from the given scenario wherever applicable.
- Use suitable diagrams, semantic representations, logical predicates, workflows, architectures, or tables whenever necessary.
- Clearly state assumptions if additional information is required.
- Duration: 30 minutes.

Section / Topic	Focus	Question No.
Representing Meaning & Meaning Structure of Language	Analyze semantic interpretation of user queries, identify ambiguity in meaning representation, evaluate semantic processing limitations, and improve understanding in banking chatbot applications.	Q1
Syntax-Driven Semantic Analysis	Analyze multiple interpretations of spoken commands, evaluate parsing-related ambiguity, and assess semantic extraction in real-time voice assistant systems.	Q2
First-Order Predicate Calculus, Representing Linguistically Relevant Concepts, Syntax-Driven Semantic Analysis	Design a semantic processing architecture for healthcare reports, model relationships among entities and actions, represent linguistic concepts, and improve semantic extraction accuracy.	Q3

Annexure A – Industry Problem Solving Task

1. AI-Powered Insurance Claim Processing System

An insurance company is developing an NLP-based system to automatically process customer insurance claims. The system uses **Context-Free Grammar (CFG)** to parse customer statements and extract information such as the claimant, damaged object, cause of damage, and claim amount.

However, the system faces the following problems:

- Ambiguous sentence structures
- Difficulty distinguishing noun and prepositional phrase attachment
- Subject–verb agreement errors
- Inefficient parsing of lengthy customer descriptions
- Difficulty handling conversational and incomplete input

Example customer statement:

"I reported the damage to the car after the accident."

The sentence can produce different interpretations depending on whether **"to the car"** is associated with *reported* or *damage*.

Tasks:

- a) Analyze the structural ambiguity in the given sentence using CFG and illustrate the possible interpretations.
- b) Evaluate the limitations of a basic CFG in handling ambiguity, agreement, and long customer-generated insurance statements.
- c) Design an improved parsing approach using **PCFG, Feature Structures, and Earley Parsing** to resolve ambiguity and efficiently process the claim.
- d) Justify how the proposed architecture can improve the accuracy, scalability, and reliability of automated insurance claim processing.

2. Intelligent Railway Ticket Booking Assistant

A railway company is developing a conversational AI assistant for ticket booking. The system currently uses **top-down parsing** to interpret passenger requests.

The system encounters:

- Excessive backtracking
- Ambiguous attachment of phrases
- Incomplete passenger requests
- Long-distance dependencies
- Slow response for complex queries

Example passenger request:

"Book two tickets from Chennai to Delhi for my parents with AC sleeper."

The system sometimes incorrectly associates **"with AC sleeper"** with the passengers instead of the ticket/class information.

Tasks:

- a) Analyze the possible syntactic interpretations of the given passenger request using CFG.
- b) Evaluate why top-down parsing may result in excessive backtracking and inefficient processing for conversational ticket-booking queries.
- c) Analyze how **Earley Parsing** can handle ambiguity, incomplete input, and different possible parse structures more effectively.
- d) Compare **Top-Down Parsing, CFG-based parsing, and Earley Parsing** in terms of ambiguity handling, computational efficiency, partial-input processing, and suitability for real-time railway booking systems.

3. AI-Based Legal Contract Analysis System

A legal technology company is developing an NLP system to analyze contracts and automatically identify:

- Parties involved
- Obligations

- Conditions
- Deadlines
- Actions
- Exceptions

The existing system uses a basic CFG parser. However, it produces incorrect interpretations when processing long legal sentences containing relative clauses, passive constructions, and multiple prepositional phrases.

Example contract sentence:

"The supplier who delivered the equipment to the company must replace the defective components within thirty days."

The system sometimes incorrectly identifies the **company** as the entity responsible for delivering the equipment.

Tasks:

- Analyze the syntactic structure of the given sentence using CFG and identify possible sources of ambiguity.
- Evaluate why basic CFG is insufficient for accurately interpreting complex legal sentences containing relative clauses and nested structures.
- Design an NLP architecture integrating **CFG, PCFG, Feature Structures, and Earley Parsing** to improve syntactic and semantic interpretation.
- Develop a step-by-step processing workflow showing how the system can extract:
 - Supplier
 - Action
 - Equipment
 - Company
 - Obligation
 - Deadline

and justify how the proposed approach improves automated contract analysis.

4. AI-Based E-Commerce Product Search System

An e-commerce company is developing a conversational product-search system. Customers enter natural-language queries rather than selecting predefined filters.

The current system uses CFG but faces difficulties with:

- Ambiguous phrase attachment
- Coordination
- Missing words
- Informal customer queries
- Long product descriptions
- Real-time search requirements

Example user query:

"Find laptops with a touchscreen for students under ₹60,000."

The system may incorrectly associate "**for students**" with the touchscreen rather than the laptop.

Tasks:

- Analyze the syntactic ambiguity in the given query and construct possible parse interpretations using CFG.
- Evaluate the limitations of CFG in handling informal, elliptical, and ambiguous e-commerce search queries.
- Propose an improved parsing architecture using **PCFG, Feature Structures, and Earley Parsing** to identify product attributes, user requirements, and price constraints.
- Justify how the proposed solution can improve search precision, response time, and customer experience in a large-scale e-commerce platform.

5. Intelligent Airline Voice Assistant

An airline is developing a real-time voice assistant that allows passengers to modify flight bookings using spoken commands.

The system currently uses a conventional top-down parser. It experiences:

- Backtracking
- Ambiguous attachment
- Speech recognition errors
- Incomplete commands
- Delayed responses
- Difficulty interpreting corrections made by users

Example spoken command:

"Change my flight to Mumbai tomorrow with two passengers."

Depending on the parse, **"with two passengers"** may be incorrectly interpreted as an attribute of the destination rather than the number of passengers.

The passenger may also say:

"Change my flight to Mumbai tomorrow... actually, make that Bangalore."

Tasks:

- Analyze the possible syntactic structures and ambiguities in the given voice command.
- Evaluate the limitations of top-down parsing when processing incomplete, corrected, and ambiguous spoken commands in real time.
- Design a parsing architecture using **Earley Parsing, PCFG, and Feature Structures** to support partial input, ambiguity resolution, and agreement constraints.
- Develop a processing workflow showing how speech input can be converted into a structured booking request and justify the proposed method for real-time airline applications.

Rubrics for Calculation

Criteria	Weight age	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Industry Problem	20%	Comprehensive understanding of real-world problem and constraints	Good understanding with minor gaps in requirements	Basic understanding; some requirements unclear	Poor understanding ; misinterprets problem context
Application of Engineering Concepts	25%	Applies advanced and relevant concepts effectively to solve the problem	Applies appropriate concepts with minor errors	Limited application of concepts	Incorrect or no application of concepts
Solution Design	25%	Innovative, practical, and feasible solution aligned with industry standards	Logical and feasible solution with minor gaps	Basic solution with limited feasibility	Unclear or impractical solution
Use of Tools	15%	Effective use of modern tools, technologies, and real data	Appropriate use of tools with correct implementation	Limited or inconsistent use of tools	Minimal or incorrect use of tools
Analysis	10%	Thorough analysis with validated results and	Good analysis with reasonable validation	Basic analysis with limited validation	Poor or no analysis

		performance evaluation			
Professionalism	5%	Highly professional, well-documented, and clearly presented work	Good documentation and presentation	Basic documentation with some gaps	Poor documentation and presentation

Q. No. / Criteria Level	Understanding of Industry Problem	Application of Engineering Concepts	Solution Design	Use of Tools	Analysis	Professionalism	Sub. Total	Total %
1	3	3	3	3	3	4	16	78%
2	3	3	3	3	3	3	15	
3	3	4	3	3	3	3	16	
4	3	3	4	3	3	3	16	
5	3	3	3	3	3	3	15	

Faculty In-charge	Course Coordinator	Program Director
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____

1. AI-Powered Insurance Claim Processing System

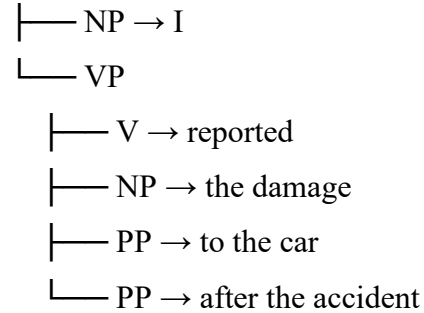
a) Analyze the structural ambiguity using CFG

Sentence:

"I reported the damage to the car after the accident."

The main structure can be represented as:

S



The ambiguity occurs because the PP "to the car" can potentially attach at different levels.

Interpretation 1: "to the car" modifies "damage"

I reported [the damage [to the car]] [after the accident].

Meaning:

The damage was damage to the car, and the person reported it after the accident.

Semantic interpretation:

Claimant → I

Action → reported

Damaged Object → car

Cause/Context → accident

Interpretation 2: "to the car" attaches to the reporting event

I [reported the damage] [to the car] [after the accident].

This would grammatically suggest that the reporting was directed "to the car", which is semantically unusual.

Therefore, although CFG may generate both structures, the first interpretation is much more plausible in the insurance domain.

CFG representation

A simplified CFG can be:

S → NP VP

VP → V NP PP PP

NP → Det N

NP → Det N PP

PP → P NP

NP → Pronoun

The important issue is that CFG mainly represents structural possibilities; it does not inherently know that "car" is a likely damaged object.

Conclusion

The most appropriate interpretation is:

I reported [damage to car] [after accident]

because "damage to the car" forms a meaningful noun phrase in an insurance context.

b) Limitations of basic CFG

A basic CFG has several limitations.

1. Structural ambiguity

A CFG may generate multiple valid parse trees.

For example:

damage [to the car]

reported [to the car]

The grammar alone cannot reliably select the intended interpretation.

2. Subject-verb agreement

Basic CFG rules such as:

$S \rightarrow NP VP$

do not automatically enforce:

He reports

They report

Feature information such as number and person is required.

3. Long sentences

Insurance customers may provide lengthy descriptions:

I was driving my car when another vehicle suddenly

hit me and damaged the front door...

The number of possible parse structures can become very large.

4. Conversational input

Real users may say:

Car damaged... accident yesterday...

or:

I want to claim for my car, actually the front door.

Basic CFG expects relatively well-formed sentences and has difficulty with fragmented input.

5. No probability

CFG treats grammatical alternatives without naturally ranking them according to likelihood.

Conclusion

Basic CFG is useful for representing syntax but is insufficient by itself for reliable real-world insurance claim understanding.

c) Improved approach using PCFG, Feature Structures and Earley Parsing

A better architecture is:

Customer Input



Text Preprocessing



Tokenization



Feature Extraction



Earley Parser



Multiple Parse Trees



PCFG Probability Ranking



Feature Structure Validation



Semantic Role Extraction



Claim Information

PCFG

A Probabilistic Context-Free Grammar assigns probabilities to grammar rules.

For example:

$$P(NP \rightarrow NP PP)$$

can be learned from training data.

If:

damage to the car

is very common in insurance claims, its parse can receive a higher probability.

Thus:

$$Parse^* = \arg \max_P P(P \mid sentence)$$

The most probable parse is selected.

Feature Structures

Feature structures can represent:

NP:

Number = Singular

Type = Vehicle

Role = DamagedObject

They can also enforce agreement:

Subject:

Number = Singular

Verb:

Number = Singular

Earley Parsing

Earley parsing is useful because it can:

- Handle general CFGs
- Handle ambiguity
- Process incomplete input
- Avoid unnecessary repeated parsing
- Work well with complex sentence structures

d) Justification of proposed architecture

Accuracy

PCFG ranks alternative parses, while feature structures eliminate linguistically invalid interpretations.

Scalability

Earley parsing can efficiently process different CFG structures and supports complex sentences.

Reliability

Semantic validation ensures that extracted information makes sense in the insurance domain.

For example:

Claimant → I

Object → car

Event → accident

Action → reported

The system can then verify:

Is "car" a possible damaged object?

Is "accident" a possible cause?

Final architecture

Input



NLP preprocessing



Earley Parser



Candidate Parses



PCFG Ranking



Feature Agreement Checking



Semantic Role Labeling



Claimant / Object / Cause / Amount



Validation



Insurance Claim Processing

Conclusion:

The combination of PCFG, Feature Structures and Earley Parsing provides better ambiguity resolution, grammatical validation, scalability and reliability than a basic CFG.

2. Intelligent Railway Ticket Booking Assistant

Sentence:

"Book two tickets from Chennai to Delhi for my parents with AC sleeper."

a) Possible syntactic interpretations using CFG

A simplified grammar:

$S \rightarrow VP$

$VP \rightarrow V NP PP PP PP$

$NP \rightarrow Det N$

$NP \rightarrow Num N$

PP $\rightarrow$ P NP

NP $\rightarrow$ Poss N

A natural interpretation is:

Book

|— two tickets

|— from Chennai

|— to Delhi

|— for my parents

|— with AC sleeper

Semantic structure:

Action $\rightarrow$ Book

Quantity $\rightarrow$ 2

Source $\rightarrow$ Chennai

Destination $\rightarrow$ Delhi

Passengers $\rightarrow$ Parents

Class $\rightarrow$ AC Sleeper

Possible incorrect attachment

The phrase:

with AC sleeper

could be attached to:

my parents

giving:

[for my parents with AC sleeper]

This is structurally possible but semantically unlikely.

Correct interpretation

[two tickets ... with AC sleeper]

Here "AC sleeper" describes the ticket/train class.

Therefore:

<i>with AC sleeper $\rightarrow$ Ticket/Class</i>
--

b) Why top-down parsing can cause excessive backtracking

Top-down parsing starts from the start symbol and tries to generate the input.

For an ambiguous sentence, it may try:

S

↓

VP

↓

NP

↓

PP

When one interpretation fails, it returns to an earlier decision and tries another.

This causes backtracking.

Problems

1. Multiple possible PP attachments
2. Long passenger requests
3. Incomplete sentences
4. Informal language
5. Repeated parsing of the same structures
6. Long-distance dependencies

Example:

Book two tickets...

The parser may not immediately know whether:

with AC sleeper

modifies:

- passengers
- tickets
- journey
- train

Therefore it may explore several alternatives.

Conversational problem

Users may say:

Two tickets Chennai Delhi...

without using a complete grammatical sentence.

A conventional top-down parser may fail because the expected grammar structure is incomplete.

c) Earley Parsing for ambiguity and incomplete input

Earley parsing uses three important operations:

1. Prediction
2. Scanning
3. Completion

It maintains states representing partially completed rules.

For example:

Book two tickets from Chennai...

Even before the entire sentence is complete, Earley parsing can maintain partial hypotheses.

Advantages

Ambiguity

It can maintain multiple possible parse structures.

Incomplete input

It can process:

Book two tickets from Chennai...

even if the user has not finished speaking.

Efficiency

It avoids restarting the entire parse for every alternative.

Conversational systems

It is useful for incremental processing.

Example:

Book two tickets...

↓

Book two tickets from Chennai...

↓

Book two tickets from Chennai to Delhi...

↓

Complete request

d) Comparison of parsing approaches

Feature	Top-Down Parsing	CFG-Based Parsing	Earley Parsing
Basic principle	Starts from S	Uses grammar rules	Dynamic-programming CFG parsing
Ambiguity	Can cause backtracking	Can produce multiple parses	Efficiently handles ambiguity
Incomplete input	Poor	Limited	Good
Backtracking	High	Depends on implementation	Reduced
Complex structures	Less suitable	Moderate	Good
Real-time	Limited	Moderate	Better

conversational input			
Long-distance dependencies	Difficult	Difficult without extensions	Better support
Probabilistic ranking	No	No in basic CFG	Can be combined with PCFG
Industry suitability	Low	Moderate	High

Conclusion

For a railway booking assistant, an Earley-based parser combined with PCFG and semantic validation is more suitable than a conventional top-down parser.

3. AI-Based Legal Contract Analysis System

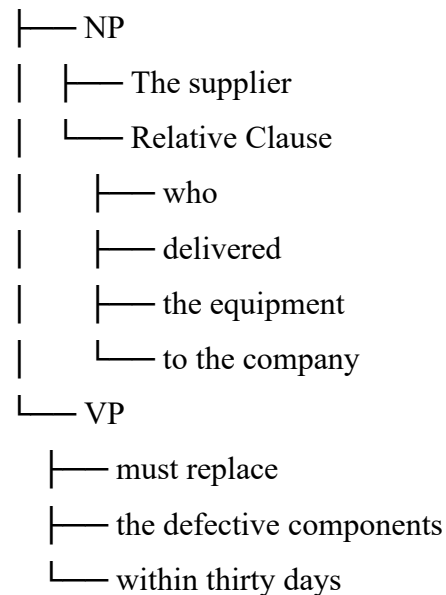
Sentence:

"The supplier who delivered the equipment to the company must replace the defective components within thirty days."

a) Syntactic structure using CFG

Main structure:

S



A simplified CFG:

$S \rightarrow NP VP$

$NP \rightarrow Det N RC$

$RC \rightarrow RelPron VP$

$VP \rightarrow V NP PP$

$VP \rightarrow Modal V NP PP$

NP → Det Adj N

PP → P NP

Semantic structure

Supplier → Agent

Delivered → Action

Equipment → Object

Company → Recipient

Supplier → Obligor

Replace → Obligation

Components → Object

Thirty days → Deadline

Potential ambiguity

The relative clause:

who delivered the equipment to the company

must be attached to "supplier".

Correct:

supplier

↓

who delivered equipment

An incorrect system could associate the delivery action with "company".

b) Why basic CFG is insufficient

1. Relative clauses

The system must understand that:

who

refers back to:

supplier

2. Nested structures

Legal sentences can contain multiple clauses:

The supplier

who delivered the equipment

to the company

must replace...

3. Semantic roles

CFG determines syntactic structure but does not inherently determine:

Supplier = Agent

Company = Recipient

4. Passive constructions

Legal documents frequently use passive voice:

The equipment was delivered by the supplier.

The grammatical subject is "equipment", but the semantic agent is "supplier".

5. Multiple PP attachments

For example:

delivered the equipment to the company

requires identifying "to the company" as the recipient/destination.

6. Long-distance relationships

The relative pronoun "who" may be separated from its semantic antecedent.

Conclusion

A basic CFG provides syntactic structure but does not provide enough probabilistic and feature-based information for reliable legal interpretation.

c) Improved NLP architecture

The proposed architecture is:

Legal Contract



Sentence Segmentation



Tokenization



POS Tagging



CFG Parsing



Earley Parsing



PCFG Ranking



Feature Structure Checking



Dependency / Semantic Role Analysis



Coreference Resolution



Legal Entity Recognition



Obligation & Deadline Extraction



Structured Legal Representation

CFG

Provides possible grammatical structures.

PCFG

Ranks alternative parses.

Feature Structures

Represent grammatical features such as:

Number = Singular

Person = Third

Tense = Past

and semantic information such as:

EntityType = Organization

Role = Agent

Earley Parser

Handles complex and ambiguous CFG structures efficiently.

d) Step-by-step extraction workflow

Sentence:

"The supplier who delivered the equipment to the company must replace the defective components within thirty days."

Step 1: Identify entities

Supplier

Equipment

Company

Components

Step 2: Identify actions

delivered

replace

Step 3: Identify semantic roles

For:

supplier delivered equipment to company

we obtain:

Agent → Supplier

Action → Delivered

Object → Equipment

Recipient → Company

Step 4: Identify obligation

The modal verb:

must

indicates an obligation.

Therefore:

Obligor → Supplier

Obligation → Replace defective components

Step 5: Identify deadline

within thirty days

Therefore:

Deadline → 30 days

Final structured representation

Supplier = Supplier

Action 1 = Delivered

Object = Equipment

Recipient = Company

Obligor = Supplier

Obligation = Replace defective components

Deadline = 30 days

Conclusion

The improved system correctly understands that the supplier delivered the equipment to the company and that the supplier has the obligation to replace the defective components within thirty days.

4. AI-Based E-Commerce Product Search System

Query:

"Find laptops with a touchscreen for students under ₹60,000."

a) Syntactic ambiguity using CFG

A simplified structure is:

S

└─ VP

- |— V → Find
- └— NP
 - |— N → laptops
 - |— PP → with a touchscreen
 - |— PP → for students
 - └— PP → under ₹60,000

Correct interpretation

Laptops

- |— touchscreen
- |— target users → students
- └— price → below ₹60,000

Semantic representation:

SEARCH(P, r, o, duct = LaptopTouchscreen = TrueUser = StudentPrice < 60000)

Possible incorrect interpretation

The parser might attach:

for students

to:

touchscreen

giving:

[touchscreen for students]

rather than:

[laptops for students]

The second is the more natural interpretation in the context of product search.

b) Limitations of CFG

Ambiguous attachment

Multiple PPs can modify different nouns.

Coordination

Queries can contain:

laptops and tablets with touchscreen

The parser must determine what the PP modifies.

Missing words

Users may type:

laptop touchscreen student 60k

There is no complete grammatical sentence.

Informal language

Users may write:

good laptop under 60k for college

Long descriptions

A customer may provide many constraints.

Real-time requirement

Large e-commerce platforms require fast parsing and search.

Therefore, a basic CFG alone is insufficient.

c) Improved architecture

User Query



Normalization



Tokenization



Entity & Attribute Extraction



Earley Parser



Candidate Parse Trees



PCFG Ranking



Feature Structure Validation



Semantic Query Representation



Product Database/Search Engine



Ranked Results

Feature structure

For the query:

Product = Laptop

Touchscreen = True

TargetUser = Student

MaximumPrice = ₹60,000

This can be represented as:

PRODUCT

|— Type: Laptop

|— Feature: Touchscreen

|— User: Student

|— Price:

|— Maximum: 60000

PCFG

PCFG selects the most likely attachment of:

for students

based on training data.

Earley Parsing

Allows multiple possible interpretations to be maintained and evaluated.

d) Benefits

Search precision

The system correctly extracts:

Laptop

Touchscreen

Student

< ₹60,000

Therefore, irrelevant products are reduced.

Response time

Earley parsing with efficient state management avoids repeatedly parsing the same structures.

Customer experience

Customers can use natural language rather than manually selecting filters.

For example:

"Find laptops for students with touchscreen

under 60k."

can directly become:

Product = Laptop

Audience = Student

Touchscreen = Yes

Price $\leq$ 60000

Conclusion

The combination of PCFG, Feature Structures and Earley Parsing improves semantic accuracy while supporting flexible natural-language product search.

5. Intelligent Airline Voice Assistant

Command:

"Change my flight to Mumbai tomorrow with two passengers."

a) Possible syntactic structures and ambiguities

A natural interpretation is:

Change

|— my flight

|— to Mumbai

|— tomorrow

|— with two passengers

Semantic representation:

Action → Change

Object → Flight

Destination → Mumbai

Date → Tomorrow

Passengers → 2

Ambiguity

The phrase:

with two passengers

could incorrectly be attached to:

Mumbai

as if it were a destination attribute.

But in the airline domain, "two passengers" is a booking constraint.

Correct:

$Passengers = 2$

Correction example

User initially says:

Change my flight to Mumbai tomorrow

Then:

actually, make that Bangalore

The system must replace:

Mumbai

with:

Bangalore

rather than treating Bangalore as a new independent request.

b) Limitations of top-down parsing

1. Backtracking

Ambiguous structures can cause repeated attempts.

2. Speech recognition errors

Speech-to-text might produce:

two passengers

as:

to passengers

The parser must use context to recover the intended meaning.

3. Incomplete commands

Users may say:

Change my flight to Mumbai...

and pause.

A rigid parser may wait for the complete sentence.

4. Corrections

Users frequently self-correct:

Mumbai... actually Bangalore.

The system must identify the correction.

5. Real-time requirement

Airline voice assistants need low latency.

Repeated parsing and backtracking can delay the response.

c) Earley + PCFG + Feature Structures

A suitable architecture is:

Speech Input



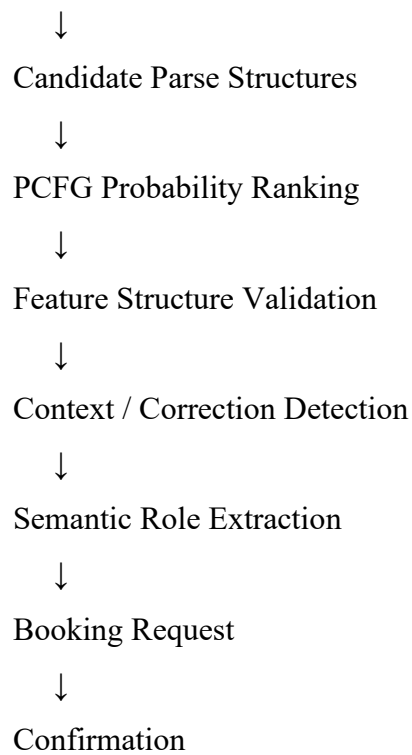
Automatic Speech Recognition



Text Normalization



Earley Parser



Earley Parser

Handles:

- Partial input
- Ambiguity
- Incremental parsing
- Complex grammar structures

PCFG

Ranks alternative interpretations.

For example:

with two passengers

is more likely to be a passenger-count constraint than a destination attribute.

Feature Structures

Represent booking information:

FLIGHT

|— Action: Change

|— Destination: Mumbai

|— Date: Tomorrow

|— Passengers: 2

After correction:

FLIGHT

|— Action: Change

|— Destination: Bangalore

|— Date: Tomorrow

|— Passengers: 2

The system updates the destination rather than creating a second booking request.

d) Complete processing workflow

Step 1: Speech input

"Change my flight to Mumbai tomorrow
with two passengers."

↓

Step 2: Speech recognition

Convert speech to text:

Change my flight to Mumbai tomorrow
with two passengers

↓

Step 3: Tokenization

Change | my | flight | to | Mumbai |
tomorrow | with | two | passengers

↓

Step 4: Earley parsing

Generate candidate syntactic structures.

↓

Step 5: PCFG ranking

Select the most probable structure.

↓

Step 6: Feature extraction

Action = Change

Destination = Mumbai

Date = Tomorrow

Passengers = 2

↓

Step 7: Correction detection

Input:

"actually, make that Bangalore"

The system detects a correction.

↓

Step 8: Update semantic representation

Destination:

Mumbai → Bangalore

↓

Step 9: Final booking request

CHANGE_FLIGHT(

 Destination=Bangalore,

 Date=Tomorrow,

 Passengers=2

)

↓

Step 10: Confirmation

The system should confirm the interpreted request before making the change.

Example:

Your request is:

Change flight to Bangalore tomorrow
for 2 passengers.

Please confirm.

This is particularly important because flight modifications can have financial consequence